Chapter 3A.



SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong

App Inventor

This set of slides is to show you a research direction (an easy-to-learn IDE for Android apps development) by MIT. You are not supposed to use App Inventor in this course.

2026-2027
COMP3330 Interactive Mobile Application Design and Programming
Dr. T.W. Chim (E-mail: twchim.cs@hku.hk)

Visual Programming Language (VPL)

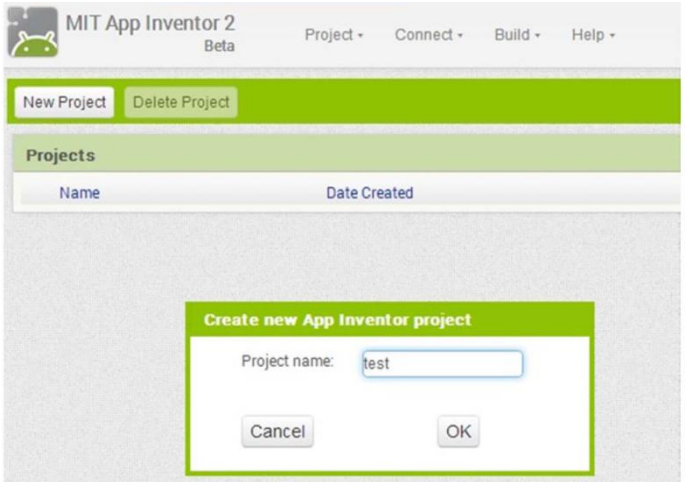
- Allows programmers to perform the coding graphically.
- The IDE provides a collection of graphical elements such as spatial texts, nodes, and icons that users assemble to program.
- Advantage:
  - Young children may find it difficult to handle complicated syntax.
  - A way to let young children understand what a program is and basic programming logics.
- A big research direction in the programming community. E.g.,
  - Choi, B., Jung, J. & Baek, Y. (2013). In what way can technology enhance student learning? : A preliminary study of technology supported learning in mathematics. In R. McBride & M. Searson (Eds.), Proceedings of Society for Information Technology & Teacher Education International Conference 2013 (pp. 3-9). Chesapeake, VA: AACE.
  - Fields, D. A., Giang, M. & Kafai, Y. B. (2013). Understanding collaborative practices in the Scratch online community: Patterns of participation among youth designers. In N. Rummel, M. Kapur, M. Nathan, & S. Puntambekar (Eds), CSCL 2013 Conference Proceedings, Volume 1. International Society of the Learning Sciences: Madison, WI, 200-207.

What is App Inventor?

- **App Inventor** (for Android) is an open-source web application originally provided by Google, and now maintained by the Massachusetts Institute of Technology (MIT).
- It allows newcomers to computer programming to create software applications for the Android operating system (OS).
- It uses a graphical interface, very similar to **Scratch** and the StarLogo TNG user interface, which allows users to drag-and-drop visual objects to create an application that can run on Android devices.
- Widely used by primary school and junior secondary school students in Hong Kong nowadays.

A simple button label example

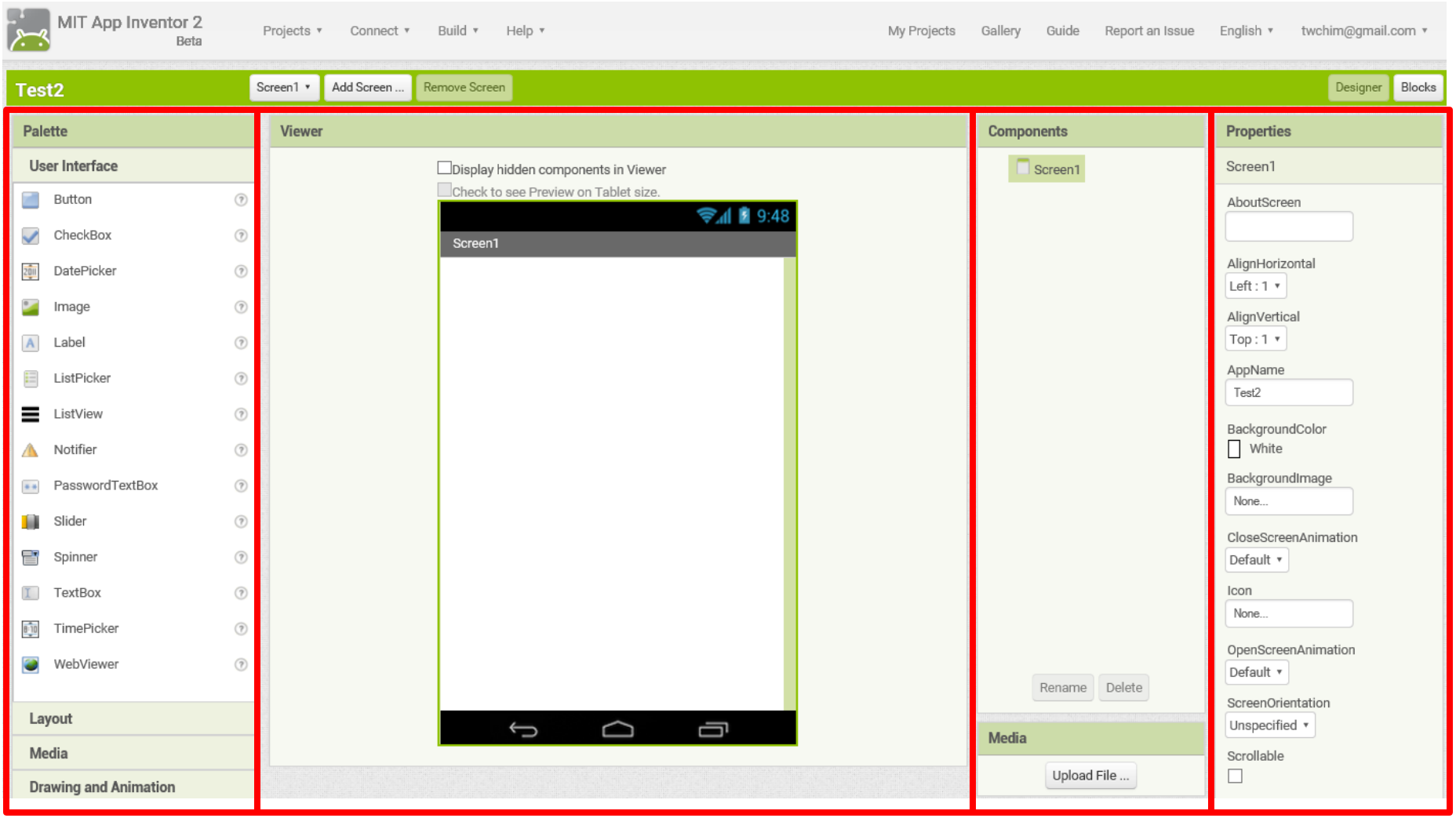
- First of all, go to the site http://ai2.appinventor.mit.edu/ and login using your Google account.
- If you haven't got a Google account, go to the site http://www.google.com/, press login and then choose "Create Account".
- Choose "New Project" to create a new project. Then type the project name and press "OK".



App Inventor working area

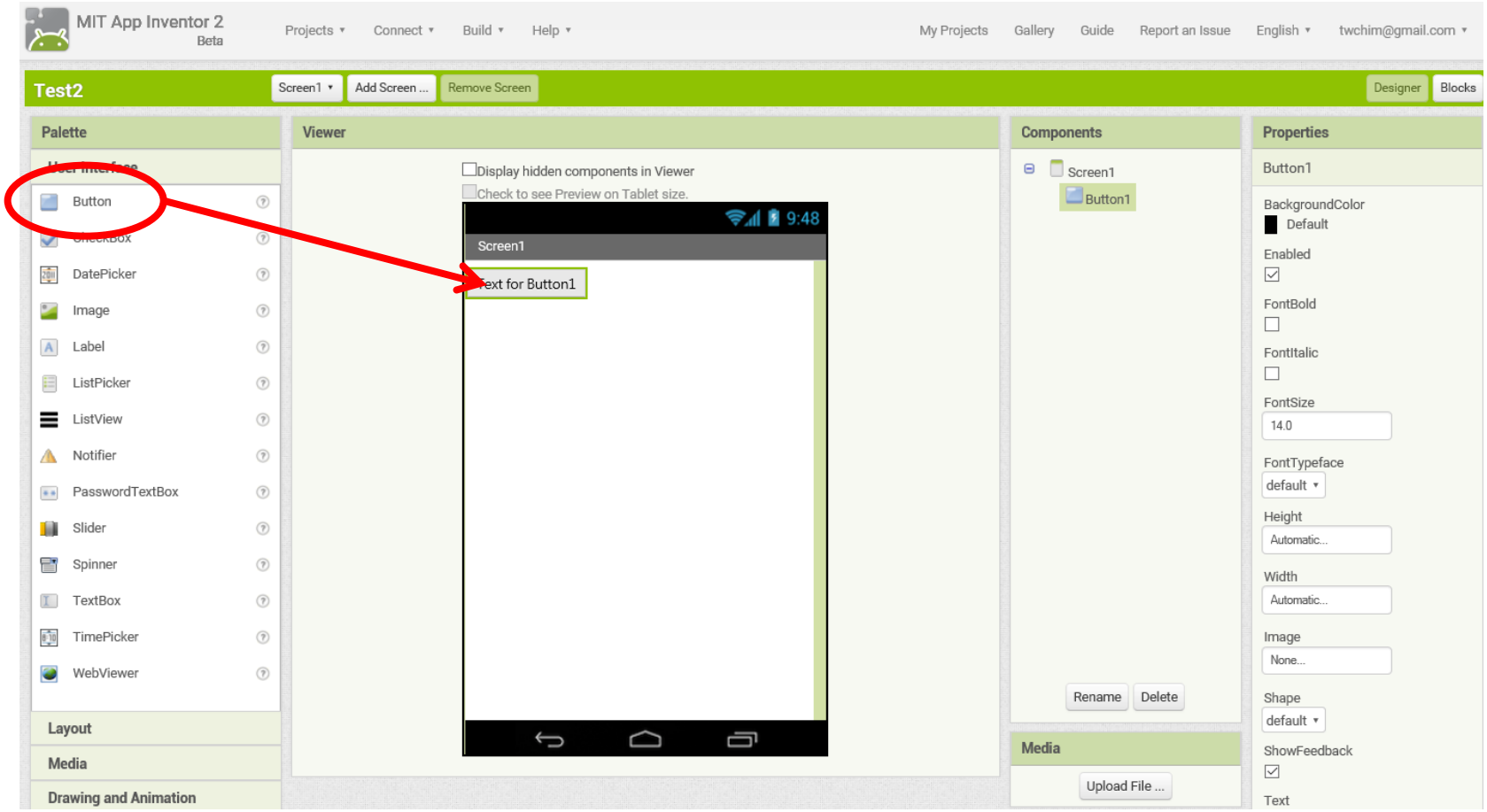
- The working area contains 4 components – Palette, Viewer, Components and Properties.
  - Palette: Materials for making your application
  - Viewer: How your application is like on the phone screen
  - Components: All components (including hidden components) in the project
  - Properties: Properties of each component (you can select and view the properties of a certain component)

A simple button label example



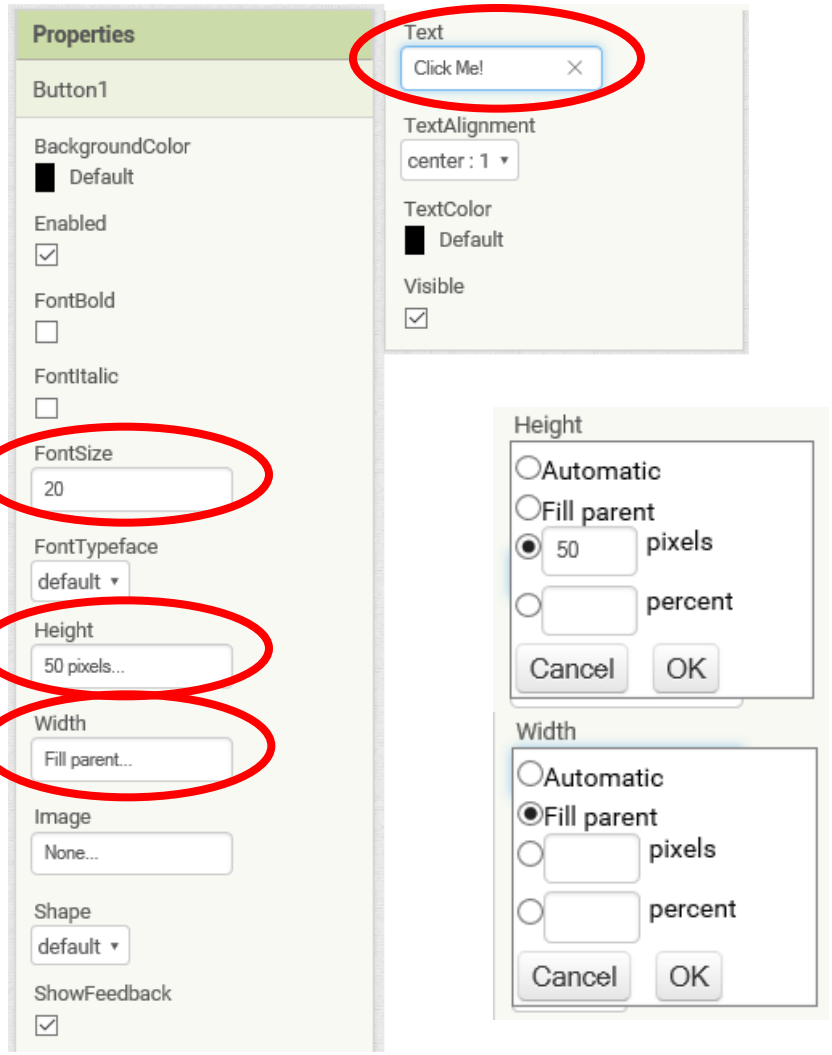
A simple button label example

- Drag a “Button” from Palette to Viewer



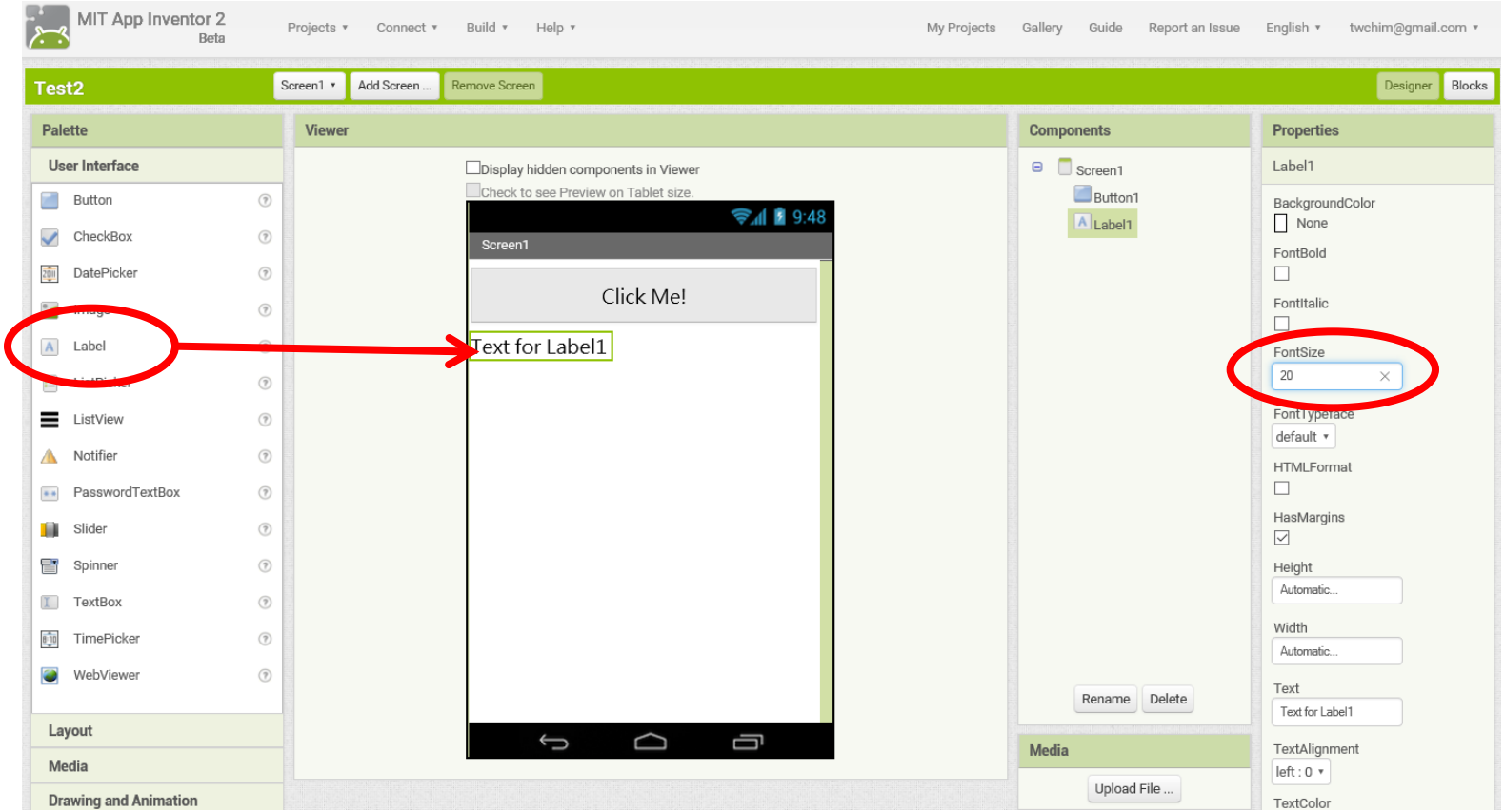
A simple button label example

- In Properties, update the following:
  - Change its “FontSize” into 20
  - Change its “Height” into 50 pixels
  - Change its “Width” into “Fill parent…”
  - Change its “Text” into “Click Me!”



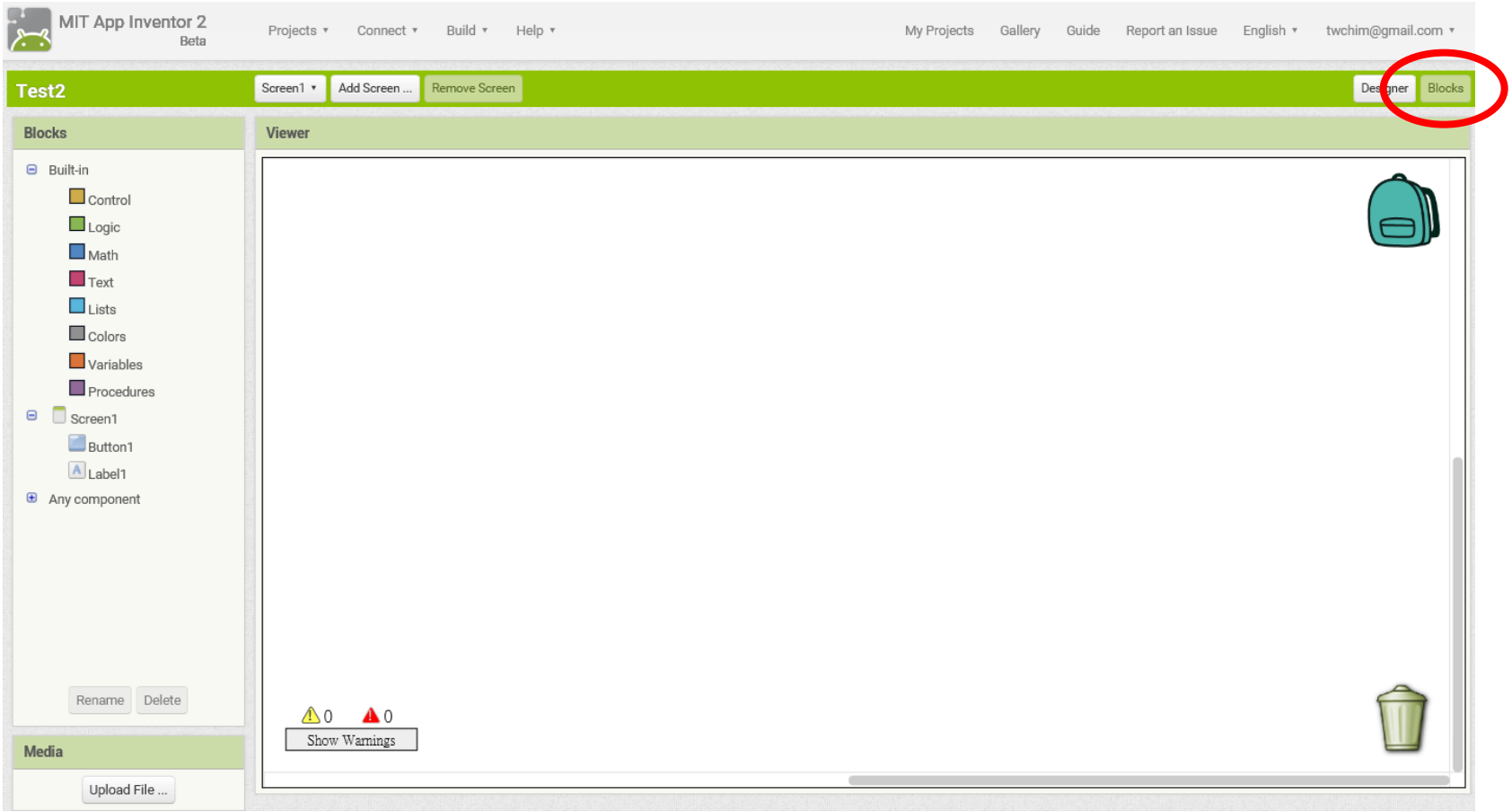
A simple button label example

- Drag a “Label” from Palette to Viewer and change its “FontSize” to 20



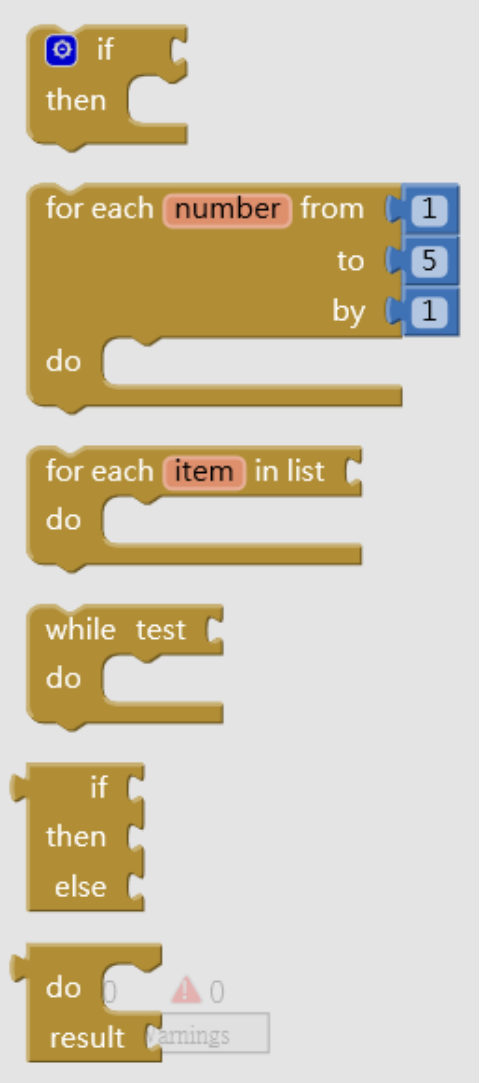
A simple button label example

- Press “Blocks” to start writing our program

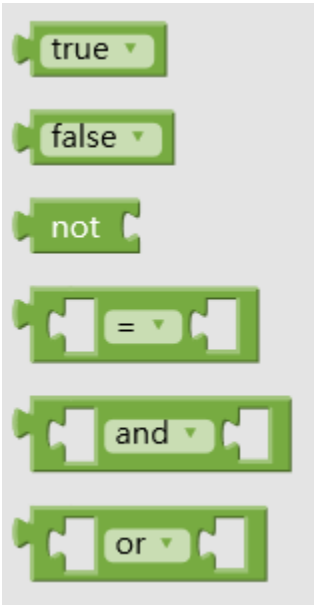


Programming Blocks Available

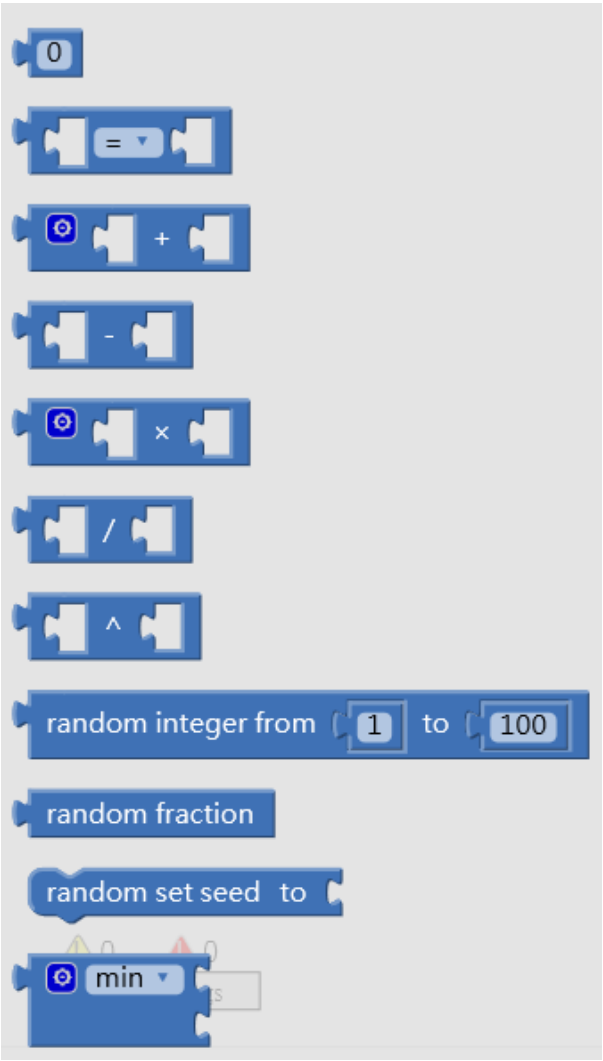
Control



Logic

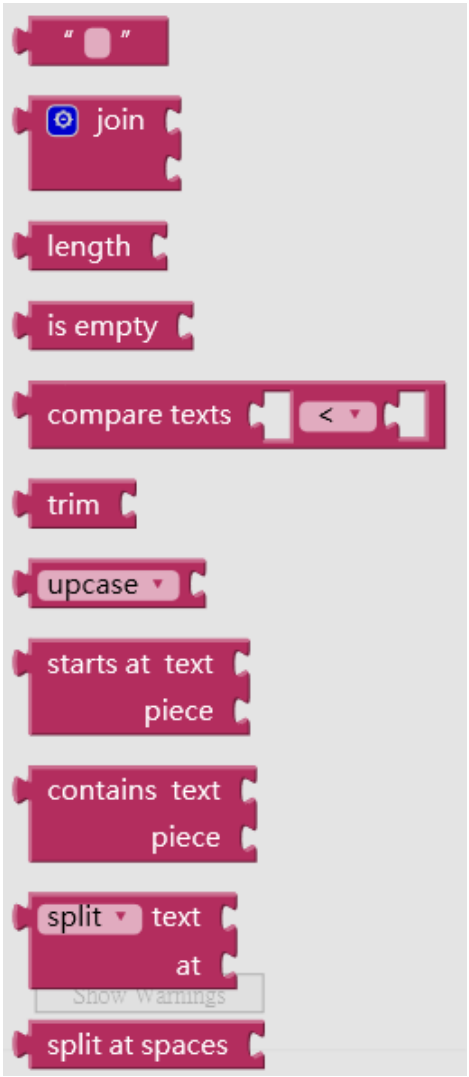


Math

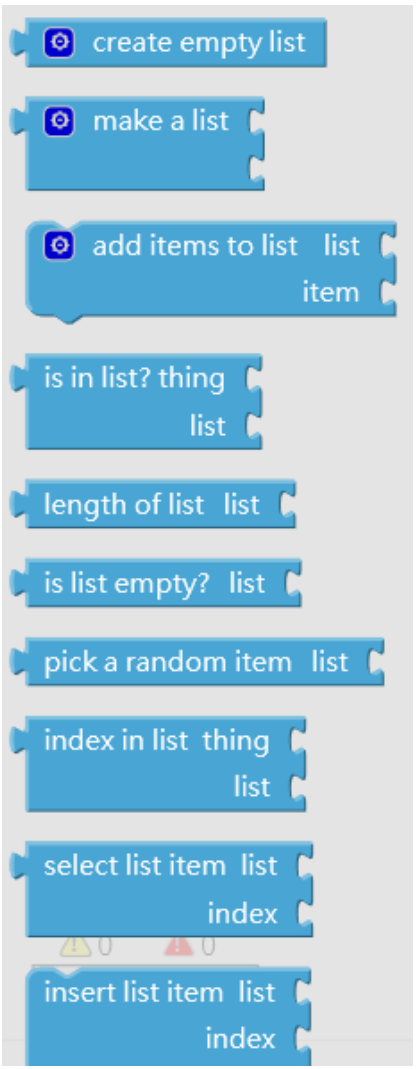


Programming Blocks Available

Text



Logic



Colors

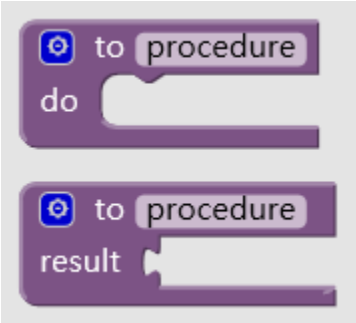


Programming Blocks Available

Variables

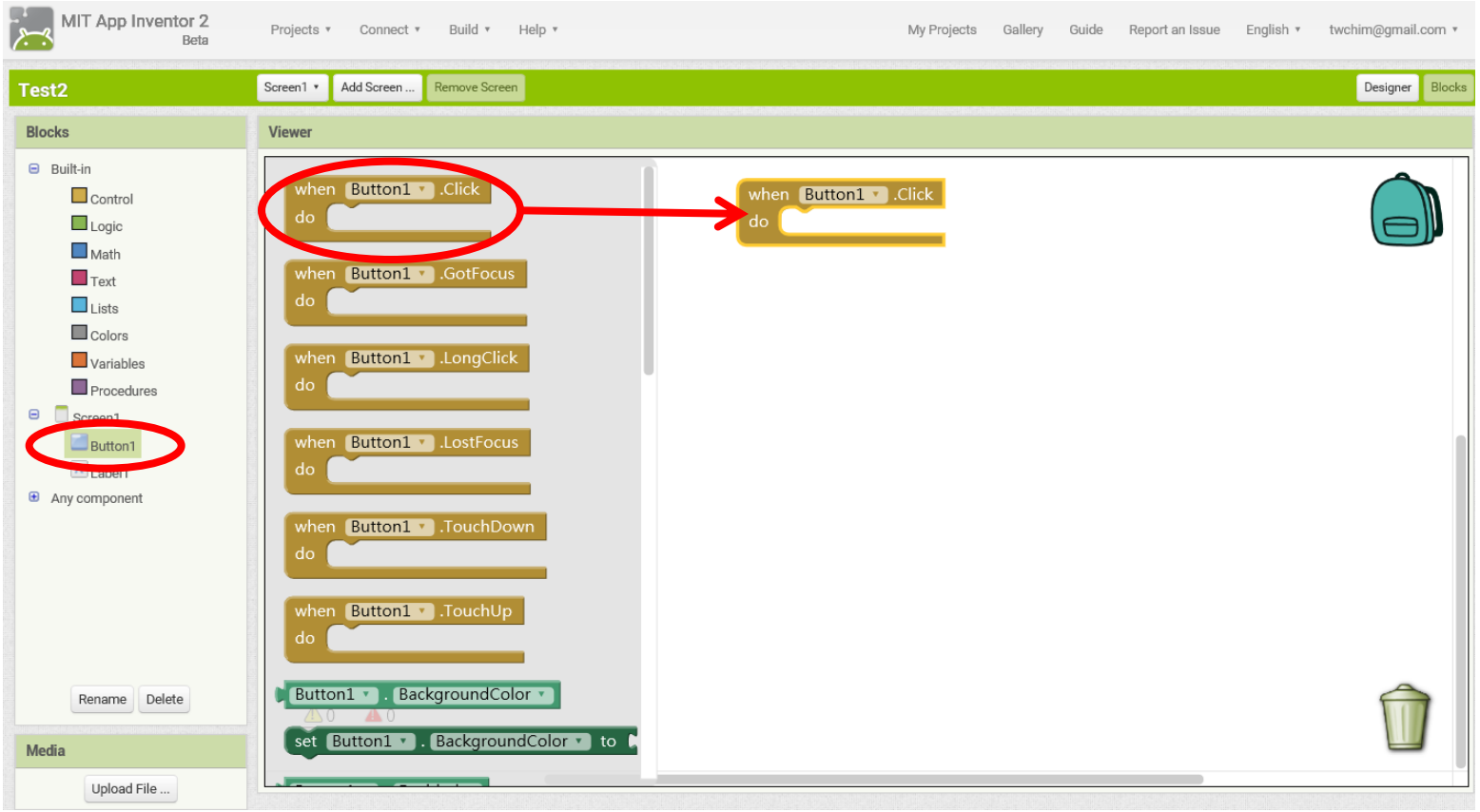


Procedures



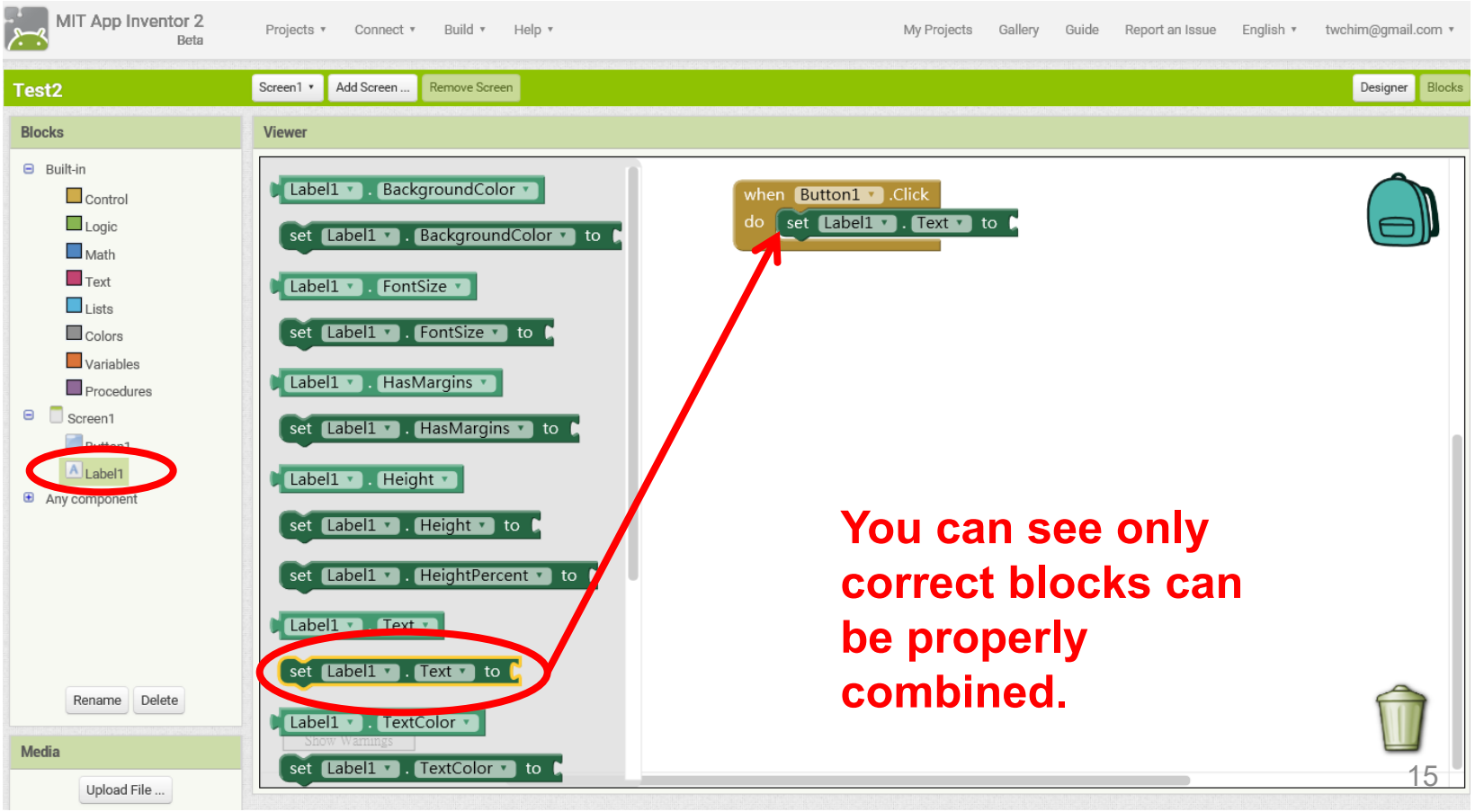
A simple button label example

- Select “Button1” and drag the first block to the right (our programming area).



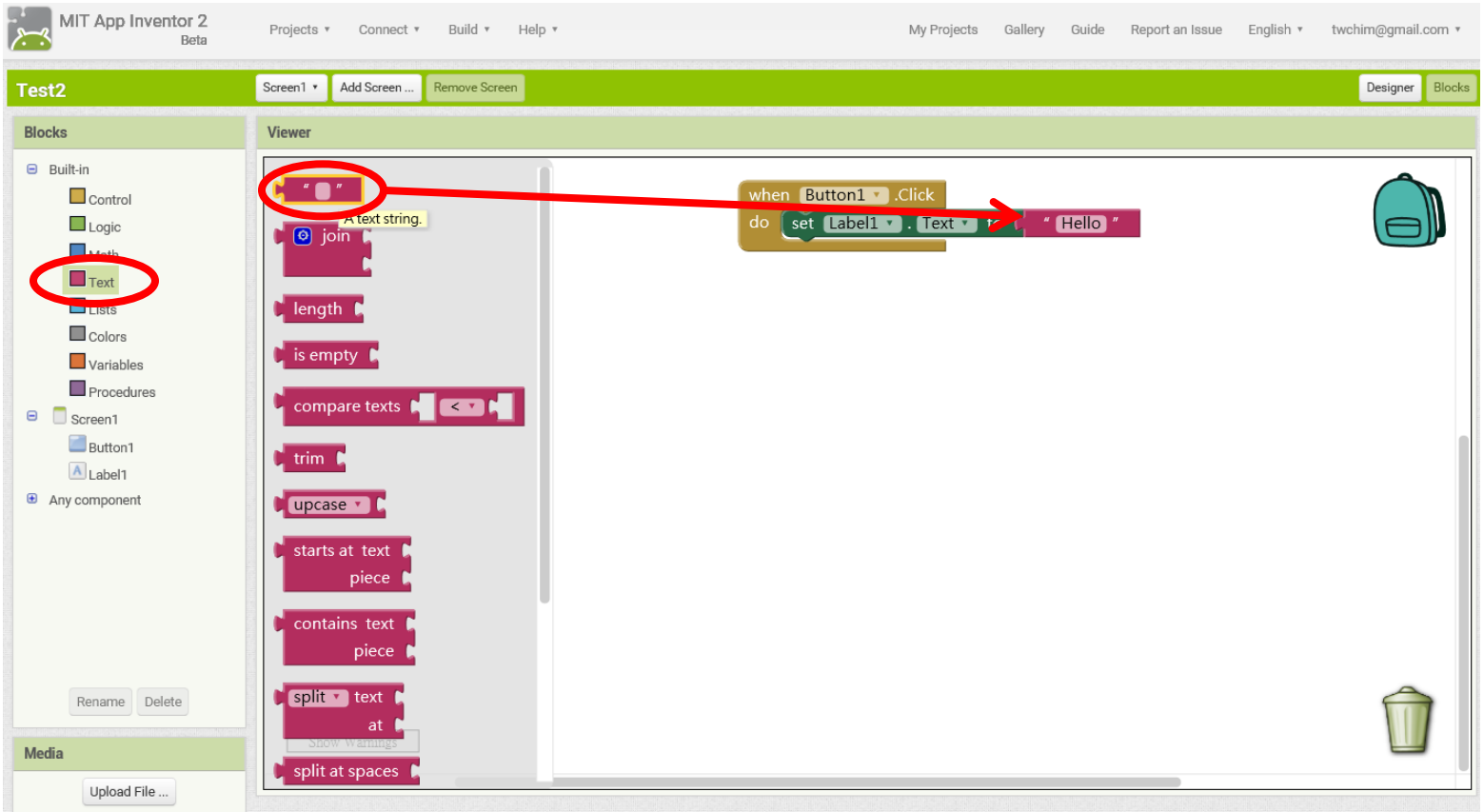
A simple button label example

- Select “Label1” and drag the block “set Label1.Text to” into the margin of the previous block.



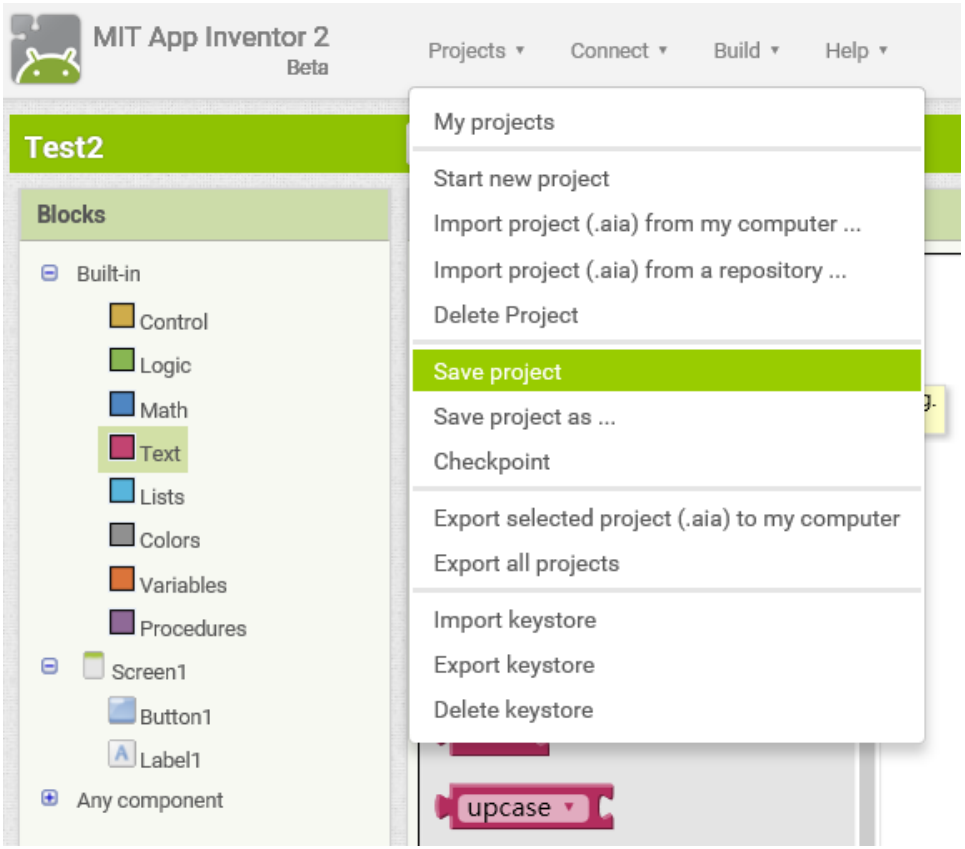
A simple button label example

- Select “Text” and drag the first block to the end of our “program”
- Change the text into “Hello”



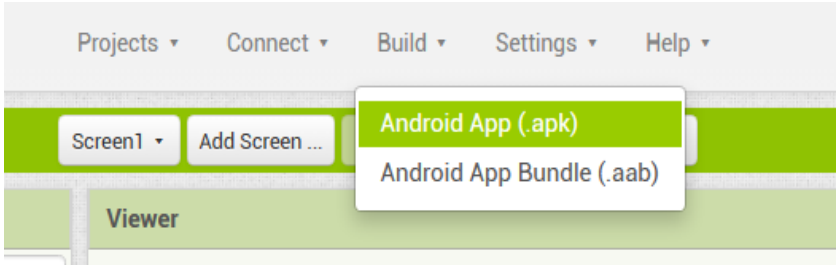
A simple button label example

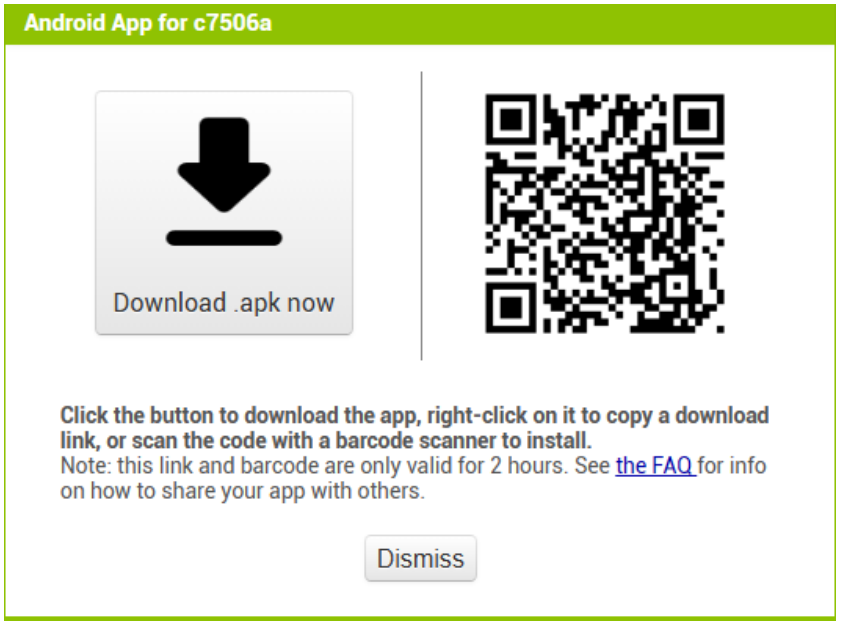
- Choose “Project” → “Save project” to save your work



A simple button label example

- Next let's try the app on your Android phone
- In the "Build" menu, choose "Android App (.apk)". You can download the .apk and transfer to your Android phone via USB cable, or use your Android phone to scan the QR code, download and install the app.





A simple button label example

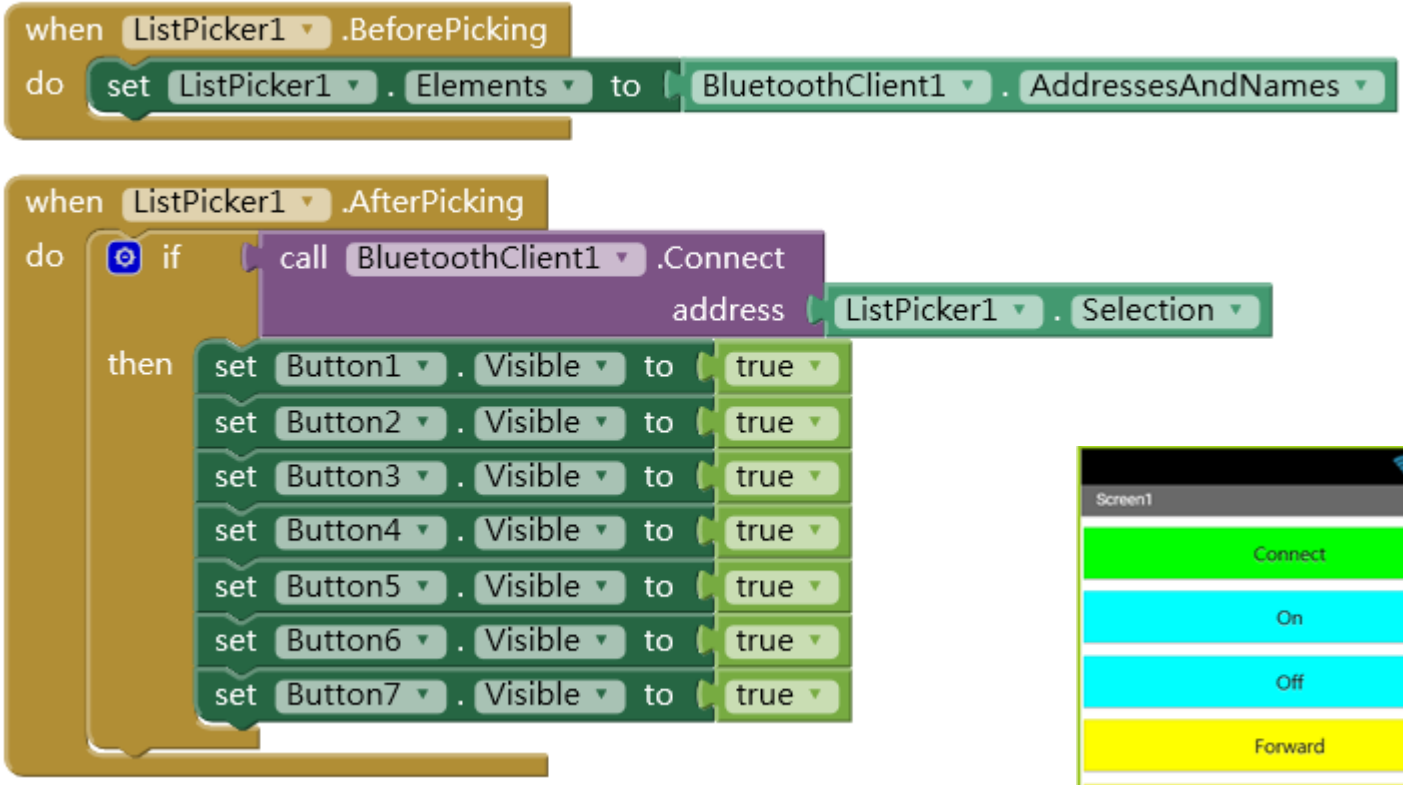
- After copying the file, your Android phone will block the installation by default.
- Go to “settings” and turn on “allow apps outside Play Store to be installed”
- In the pop-up box, choose “only allow this installation” and click “OK” to confirm
- Choose “Install” to start the installation

Try the app and you have done your first Android app!!!

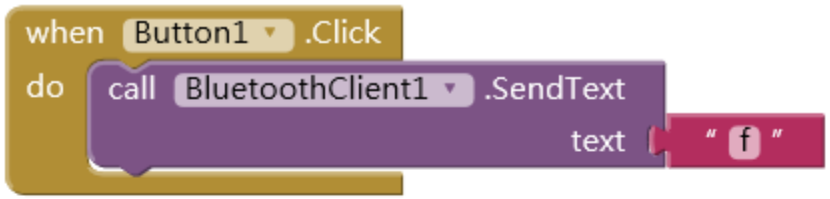


Advanced applications

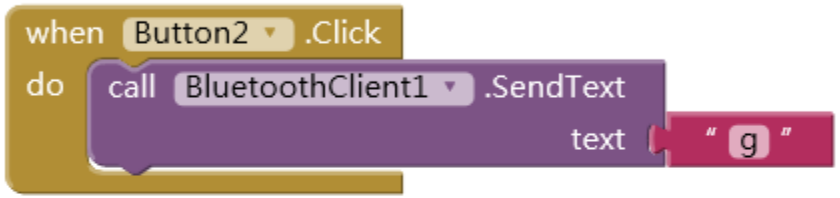
- Although App Inventor is targeted at beginners, it can be used to develop more complicated and advanced applications.

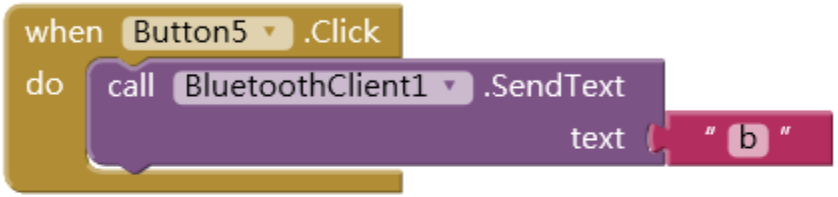


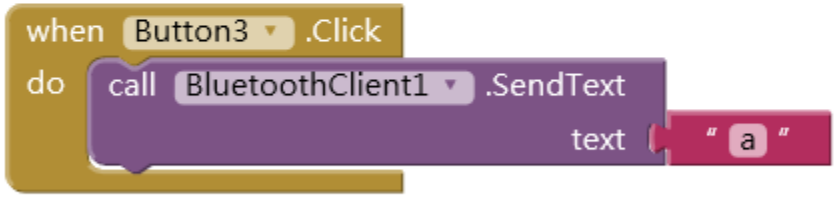
Advanced applications

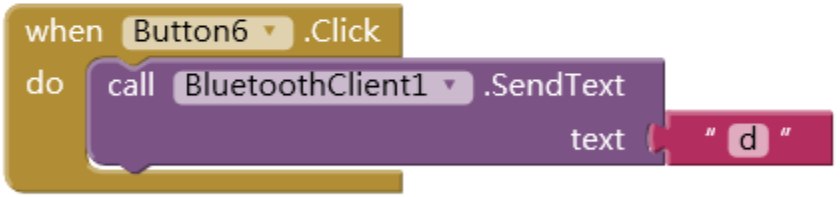








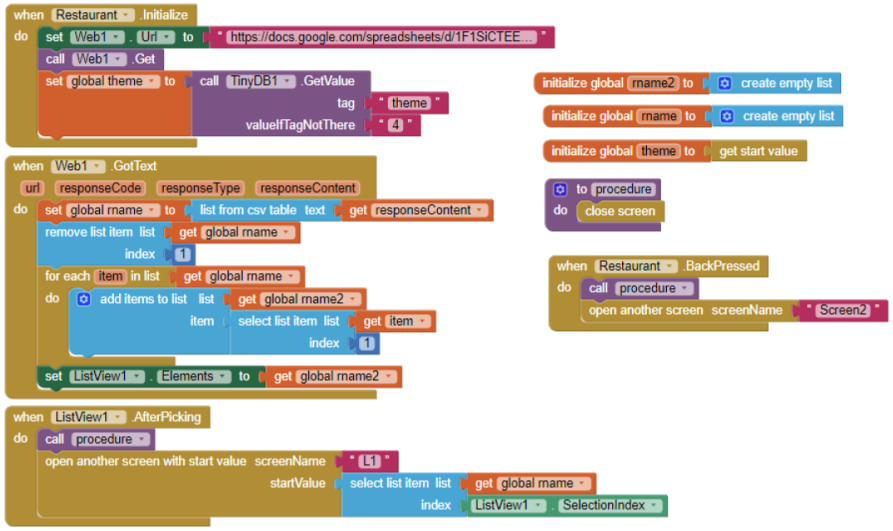




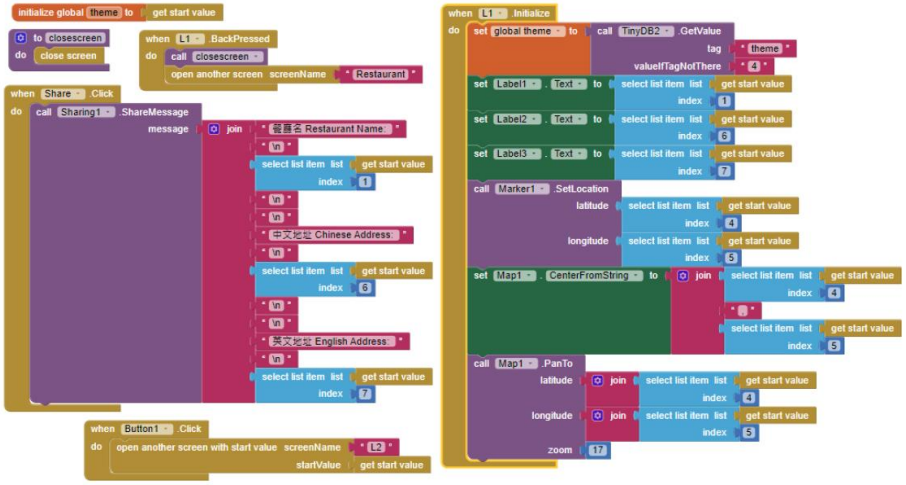


Sample: A MSc Project in 2020-2021

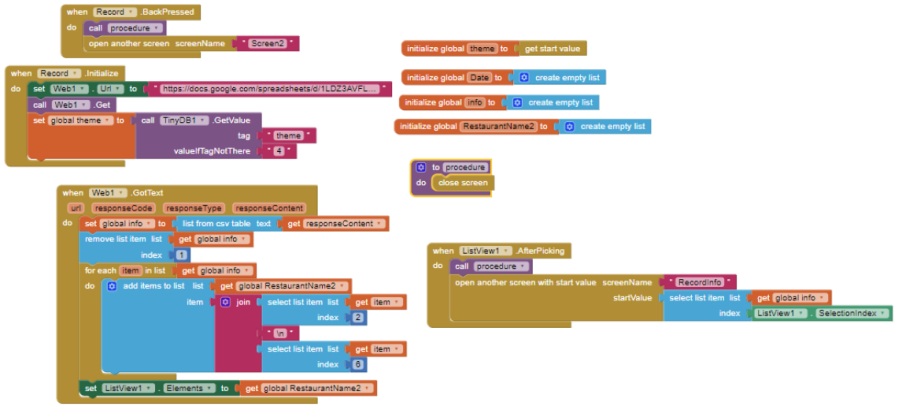
Restaurant Page



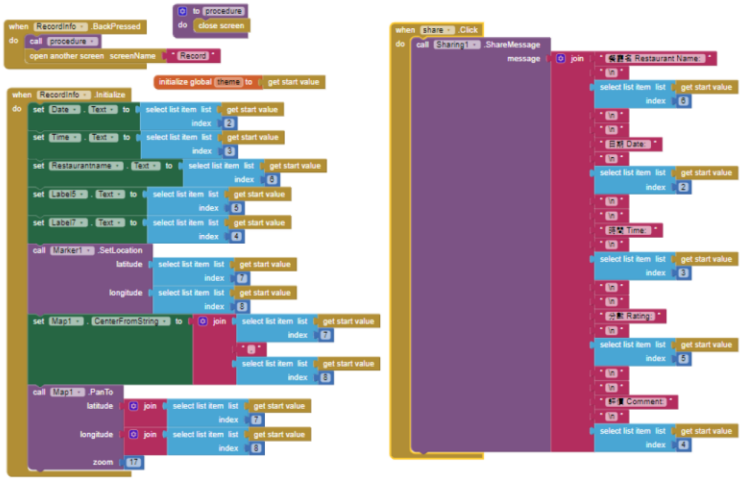
Restaurant Information Page



User Record Page



Record Information Page



Chapter 3A.

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End

